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B.Tech. Degree V Semester Supplementary Examination April 2019

IT 15-1504 DESIGN AND ANALYSIS OF ALGORITHMS (2015 Scheme)

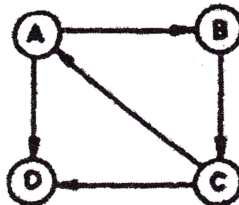
Time : 3 Hours

Maximum Marks : 60

PART A (Answer ALL questions)

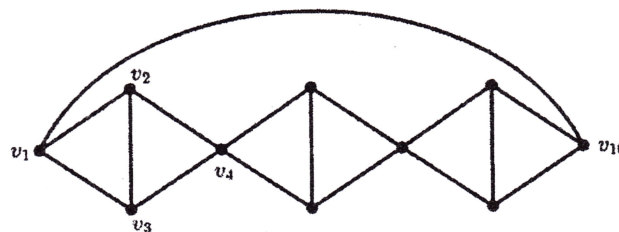
(10 × 2 = 20)

- I. (a) What is the need for analyzing an algorithm? What do you mean by asymptotic growth rate?
- (b) Write about the dynamic programming algorithm design strategy. Give example.
- (c) Compare the best case and worst case time complexity of Insertion Sort, Quick Sort, Merge Sort and Heap Sort.
- (d) Define amortized time analysis.
- (e) Consider the graph shown in the figure below:



Identify the strongly connected components in the graph.

- (f) Construct an optimal binary search tree for the following data.
Keys = {10, 12, 20}
Frequency = {34, 8, 50}
- (g) Write a note on approximation algorithms.
- (h) What is the chromatic number for the following graph? Show the solution.



- (i) State transitive closure of a binary relation.
- (j) List the properties of Red-Black trees.

PART B

(4 × 10 = 40)

- II. (a) Solve using Master's Theorem (5)

$$T(n) = 2T(n/4) + n^{0.51}$$
- (b) Solve using Substitution Method (5)

$$T(n) = 8T(n/2) + n^2 \text{ given } T(1) = 1$$

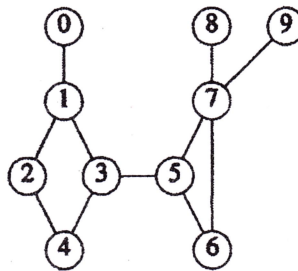
OR

(P.T.O.)

- III. (a) Write in detail about the various asymptotic notations used for representing the asymptotic growth rate. (8)
- (b) Discuss the Backtracking programming strategy with an example. (2)
- IV. Write the quicksort algorithm and derive the analysis for the best case and worst case complexity. (10)

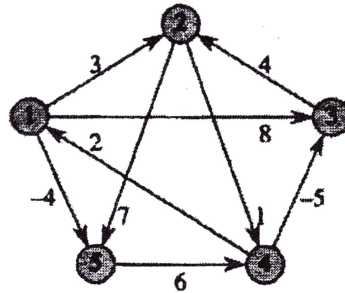
OR

- V. (a) Compare Binomial and Fibonacci Heap. (5)
- (b) Construct a RED-BLACK tree for values 2, 1, 4, 5, 9, 3, 6, 7. The values should be inserted in the given order. Now perform deletion of values 5, 9, 2. (5)
- VI. (a) Compare DFS and BFS traversal. (4)
- (b) Write the algorithm to identify biconnected components of an undirected graph. Explain the process of identifying biconnected components from the following graph. (6)



OR

- VII. Write the algorithm to find all pair shortest path and also identify the same using the algorithm for the following figure. All stages should be clearly shown. (10)



- VIII. (a) Compare P and NP-class problems. (4)
- (b) What is bin packing problem? Discuss 3 strategies to solve the bin packing problem using an example. (6)

OR

- IX. Write two algorithms for solving Travelling Salesman Problem. Apply any one strategies in the following graph to find the TSP solution. (10)

